Part 1: Approach and Main Steps

Group background and use of Codex

Our group mainly has a Media Technology background, and we had not previously built a machine learning project from scratch. For Problem 1, we used Codex to help write and debug the code and to explain concepts we encountered along the way, including Flow Matching, projection, feasibility and fidelity. This support helped us work through the implementation while learning the ideas behind each step.

Aim and setup

The aim was to test whether stronger or earlier constraints improve generated samples. We used a torus with radii $R = 3,\ r = 1$, as required in the project. We compared four sampling methods using the same trained Flow Matching model. The key distinction was between feasibility (being on the surface) and fidelity (matching the target distribution).

1. Generate the data

We generated 20,000 training points and an independent set of 4,000 reference points. We chose a uniform distribution over surface area, since the project slides did not specify a sampling density. The angle around the main ring is uniform. The tube angle ϕ has density $p(\phi) = \frac{R + r\cos\phi}{2\pi R}$, sampled by rejection sampling. This accounts for the larger area on the outside of the torus.

2. Define the constraint and projection

For a point (x, y, z), we calculated $\rho = \sqrt{x^2 + y^2}$ and $q = \sqrt{(\rho - R)^2 + z^2}$. Its distance to the surface is $|q - r|$. Projection first finds the nearest center on the main ring and then scales the offset to length r. We checked that projection returns points on the surface, leaves already projected points unchanged, and handles special points without division by zero.

3. Train one Flow Matching model

We sampled Gaussian noise x_0, a data point x_1 and a time t between 0 and 1. The input was $x_t = (1 - t)x_0 + tx_1$, and the target velocity was $x_1 - x_0$. A small neural network with three hidden layers of 128 units and SiLU activations learned this velocity using mean squared error. We trained it with Adam for 6,000 steps, batch size 512 and learning rate 0.001. The training seed was 42.

4. Compare four sampling methods

All methods started from the same noise and used 100 Euler steps. A generated freely. B moved each intermediate point partway toward its projection, using $\alpha = 1 - \exp(-5/100)$. C projected after every step. D generated freely and projected only once at the end. No method required a separate model or an extra training loss.

5. Evaluate and interpret

We measured surface distances and the fraction within 0.01 of the surface. We measured fidelity with 3D sliced Wasserstein-1 over 128 directions and total variation (TV) of a 12 by 12 joint-angle histogram against the exact target. We also used two independent true samples as a finite-sample baseline, repeated sampling with three extra seeds, and checked soft strengths and Euler step counts. Both hard methods achieved 100% feasibility, but terminal projection had lower angular TV (0.0855) than repeated projection (0.4725). These results show that feasibility does not guarantee fidelity. The experiment used one trained model, so the results are not a general ranking for every application.